CSC-343: Midterm Exam

(First-Order Logic Review Questions)

- **Show all work** for full credit. Answers without supporting steps or reasoning will receive little or no credit.
- Present your work in a **clear, logical, and organized manner**. Do not scatter steps with arrows or disconnected notes.
- **Clearly indicate your final answers** (e.g., box or highlight them).
- For proofs or problems, state all assumptions explicitly and justify each step.
- Only non-graphing calculators are permitted. No other electronic devices (including cell phones) are allowed.
- Use a pencil or erasable pen for your responses.
- For questions requiring responses that are not computational, you must use complete sentences.
- Any extraneous commentary about exam content will result in a reduced professionalism grade. If you have concerns, ask clarifying questions instead.
- Scratch paper is for your use only. It will be collected but not reviewed. Transfer all final work to this exam.
- Your responses should reflect the notation and techniques presented in class. While alternative conventions exist, consistency with course material is required for full credit.

By signing below, I certify that the work on this exam is my own without the use of external resources.

_____ _____

Printed Name Signature

For first-order logic (FOL) questions, variables are denoted by uppercase letters (X, Y, Z); constants are lowercase (a, b, c) or proper nouns.

Short Answer

1. What are some key limitations of propositional logic that motivated the development of FOL?

2. Explain the difference between a term and a sentence in FOL. Give an example of each.

3. Explain the unification algorithm and its role in making resolution inference work in FOL.

4. What is the "occurs check" in the unification algorithm? Give an example of when it prevents unification.

5. What is Universal Instantiation, and how does it enable reasoning with universally quantified statements?

6. What role does backtracking play in Prolog's execution model, and how does it relate to exploring the search space?

7. What is Generalized Modus Ponens, and how does it combine Universal Instantiation with standard Modus Ponens?

True or **False**

1. In FOL, the expression `Father(John)` could be either a term or a sentence depending on how `Father` is defined in the knowledge base.

2. The sentence $\forall X \exists Y\ Loves(X, Y)$ has the same meaning as $\exists Y \forall X\ Loves(X, Y)$.

3. Unification of the expressions $P(X, a)$ and $P(b, X)$ succeeds with the substitution $\{X/a, X/b\}$.

4. In FOL, $\forall X\ (Student(X)\ \wedge\ Smart(X))$ means "All students are smart."

5. The occurs check in the unification algorithm prevents the creation of infinite term structures.

6. Forward chaining in FOL is goal-directed, meaning it only derives facts relevant to the query.

7. Universal instantiation allows us to substitute any ground term for a universally quantified variable.

8. The expression $\forall x \sim P(x)$ is logically equivalent to $\sim \exists x\ P(x)$.

Other Questions

1. For each pair of expressions, determine whether they can be unified. If yes, provide the most general unifier (MGU). If no, explain why unification fails.

		Substitution
`Likes(X, Ice_Cream)`	`Likes(John, Y)`	
`Parent(X, Father(X))`	`Parent(Alice, Y)`	
`Knows(Father(X), X)`	`Knows(Father(Y), Mother(Y))`	
`P(X, f(X), g(Y))`	`P(A, f(A), Z)`	
`Ancestor(Tom, Father(Tom))`	`Ancestor(X, X)`	
`F(X, g(X))`	`F(g(Y), Y)`	

2. Given the following knowledge base, use forward chaining with Generalized Modus Ponens to derive all possible new facts. Show each step including the unification and the new fact derived.

Facts	**Rules**
`Bird(Tweety)`	`∀x Bird(X) → Has_Feathers(X)`
`Bird(Polly)`	`∀x Penguin(X) → Bird(X)`
`Penguin(Pete)`	`∀x (Bird(x) ∧ ~Broken_Wing(x)) → Can_Fly(x)`
`Broken_Wing(Tweety)`	`∀x Has_Feathers(x) → Warm(x)`

3. For each attempted unification below, determine if it fails due to the occurs check. If it does fail, explain why. If it succeeds, give the substitution. **Note:** The notation `{Y/X}` means "substitute `X` for `Y`" (i.e., replace all occurrences of `Y` with `X`).

(a) Unify: `X` with `f(Y)` where the current substitution is `{Y/X}`.
(b) Unify: `X` with `g(A, X)`.
(c) Unify: `f(X, Y)` with `f(g(Y), A)`.
(d) Unify: `X` with `h(Y, Z)` where the current substitution is `{Z/f(X)}`.

4. Consider the following Prolog knowledge base that defines `append` exactly as we saw in class:

```
append([], L, L).

append([H|T1], L2, [H|T3]) :- append(T1, L2, T3).
```

(a) Trace the execution of the query `append([1, 2], [3, 4], Result)` showing how the result is constructed.

(b) Trace the execution of the query `append(X, Y, [1, 2])` showing all solutions found through backtracking.